

Vorlesung 5

Eindeutigkeit der Lösung der Poissonschen Gleichung mit Dirichletscher und Neumannsche Randbedingungen

$$\Delta \Phi(\vec{r}) = - \frac{\rho(\vec{r})}{\epsilon_0} \quad | \quad + \text{ D. oder N. Randb.}$$

Annahme: zwei verschiedene Lösungen $\Phi_1(\vec{r})$ und $\Phi_2(\vec{r})$

Differenz: $u(\vec{r}) \equiv \Phi_2(\vec{r}) - \Phi_1(\vec{r}) \Rightarrow \Delta u = 0$

Dirichl.: $u(\vec{r})|_S = 0$
Neumann: $\frac{\partial u}{\partial n}|_S = 0$

1. Greensche Identität: $\int_V (\vec{\nabla} \varphi \cdot \vec{\nabla} \psi + \varphi \Delta \psi) d^3r = \oint_S \varphi \frac{\partial \psi}{\partial n} da$
(Gl. (*) in Vorl. 4)

$$\left. \begin{array}{l} \varphi = u \\ \psi = u \end{array} \right] \Rightarrow \int_V (|\vec{\nabla} u|^2 + u \Delta u) d^3r = \oint_S u \frac{\partial u}{\partial n} da$$

$$\int_V |\vec{\nabla} u|^2 d^3r = 0$$

$$\vec{\nabla} u = 0 \Rightarrow u = \text{konst}$$

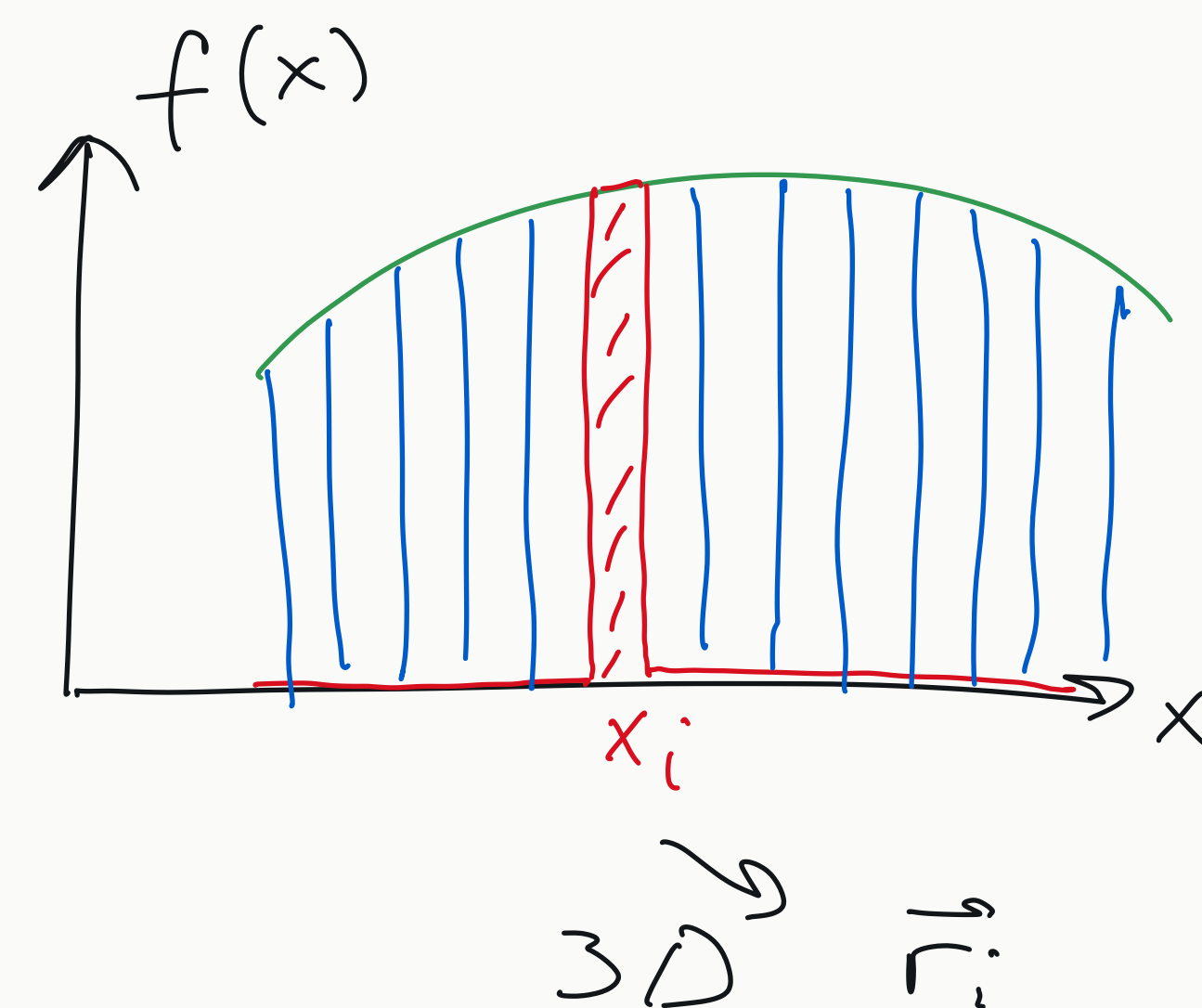
Dirichlet: $u|_S = 0$

$\text{konst} = 0 \Rightarrow \boxed{\Phi_1(\vec{r}) = \Phi_2(\vec{r})}$

Neumann: unwichtig
 $\Phi_2(\vec{r}) = \Phi_1(\vec{r}) + \text{konst}$

Greensche Funktion

Die Idee $\Delta \Phi = f(\vec{r})$

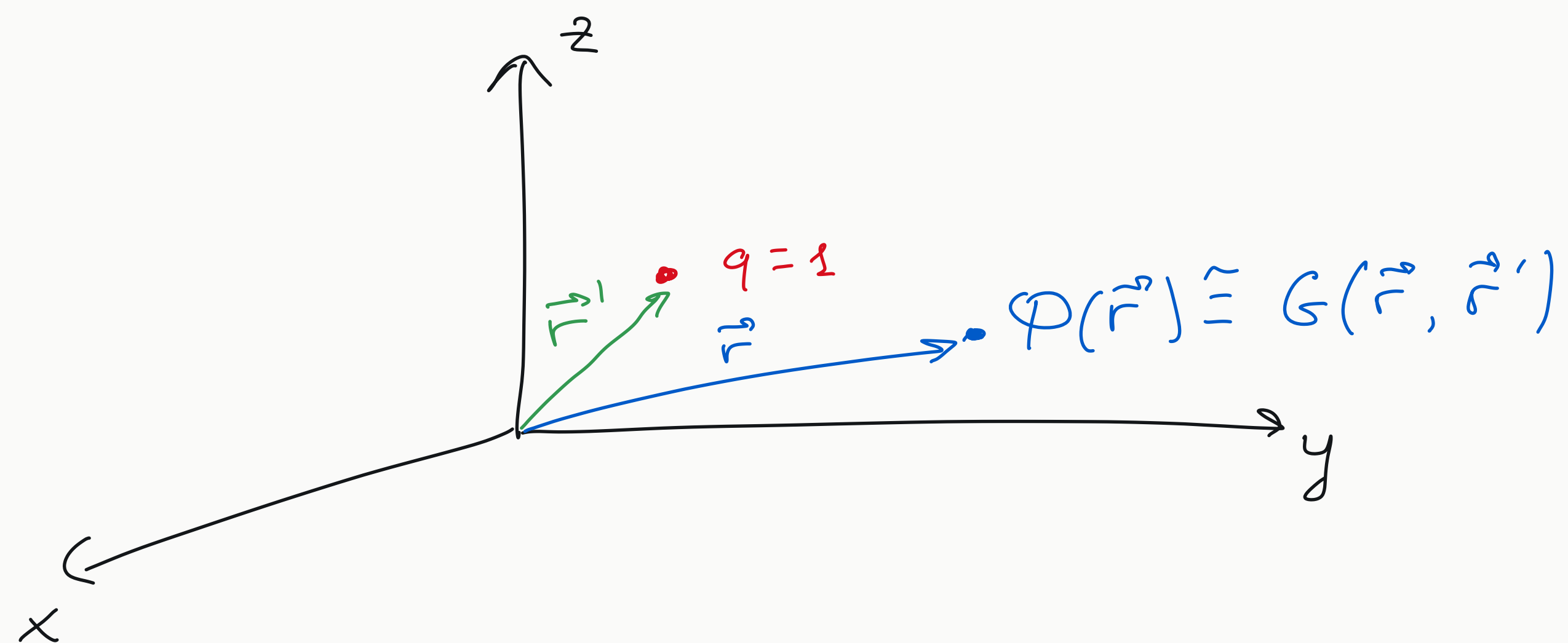


$$f(\vec{r}) = \sum_i f(\vec{r}_i) \underbrace{\delta(\vec{r} - \vec{r}_i)}_{\text{Lösung}} \Delta r^3 = \int_{\vec{r}_i \rightarrow \vec{r}'} f(\vec{r}') \delta(\vec{r} - \vec{r}') d^3 r'$$

↓ Superpositionsprinzip

$$\boxed{\Phi(\vec{r}) = \sum_i f(\vec{r}_i) \Delta r^3 \underbrace{G(\vec{r}, \vec{r}_i)}_{\text{Greensche Funktion}} = \int_{\vec{r}_i \rightarrow \vec{r}'} f(\vec{r}') \underbrace{G(\vec{r}, \vec{r}')}_{\text{Greensche Funktion}} d^3 r'}$$

Greensche Funktion



Freie Randbedingung: $\Phi(r=\infty) = 0$

Allgemeine Definition:

$$\Delta G(\vec{r}, \vec{r}') = \delta(\vec{r} - \vec{r}')$$

$$G(\vec{r}, \vec{r}') = -\frac{1}{4\pi} \frac{1}{|\vec{r} - \vec{r}'|}$$

Jackson - 1.10

$$\Delta' G(\vec{r}, \vec{r}') = \underline{\underline{-4\pi \delta(\vec{r} - \vec{r}')}}$$

$$G(\vec{r}, \vec{r}') = \frac{1}{|\vec{r} - \vec{r}'|}$$

Dieser Vorlesung

$$\Delta' G(\vec{r}, \vec{r}') = \underline{\underline{-4\pi \delta(\vec{r} - \vec{r}')}}$$

$$G(\vec{r}, \vec{r}') = \frac{1}{|\vec{r} - \vec{r}'|}$$

Allgemeine Randbedingungen: $G(\vec{r}, \vec{r}') = \frac{1}{|\vec{r} - \vec{r}'|} + \underbrace{F(\vec{r}, \vec{r}')}_{\text{Lösung der Lapl. Gl.}}$

Lösung der Lapl. Gl.

$$\Delta F(\vec{r}, \vec{r}') = 0$$

Green'scher Satz

$$\psi(\vec{r}) = \Phi(\vec{r})$$

$$\psi(\vec{r}) = G(\vec{r}, \vec{r}')$$

$$\Delta \Phi(\vec{r}) = -\frac{\rho(\vec{r})}{\epsilon_0}$$

$$\Delta G(\vec{r}) = -4\pi \delta(\vec{r} - \vec{r}')$$

$$\int_V [\Phi(\vec{r}) \Delta G(\vec{r}, \vec{r}') - G(\vec{r}, \vec{r}') \Delta \Phi(\vec{r})] d^3r = \oint_S \left[\Phi(\vec{r}) \frac{\partial G(\vec{r}, \vec{r}')}{\partial n} - G(\vec{r}, \vec{r}') \frac{\partial \Phi(\vec{r})}{\partial n} \right] da$$

$$-4\pi \Phi(\vec{r}') - \int_V G(\vec{r}, \vec{r}') \left(-\frac{\rho(\vec{r})}{\epsilon_0} \right) d^3r = \oint_S \dots$$

$$\vec{r}' \leftrightarrow \vec{r}$$

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V G(\vec{r}', \vec{r}) \rho(\vec{r}') d^3r' + \frac{1}{4\pi} \oint_S G(\vec{r}', \vec{r}) \frac{\partial \Phi}{\partial n'} - \Phi(\vec{r}') \frac{\partial G(\vec{r}', \vec{r})}{\partial n'} da'$$

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V G(\vec{r}', \vec{r}) \rho(\vec{r}') d^3r' - \frac{1}{4\pi} \oint_S \Phi(\vec{r}') \frac{\partial G(\vec{r}', \vec{r})}{\partial n'} da'$$

$G(\vec{r}, \vec{r}')|_S = 0$
unsere Wahl

Dirichletsche
Randb.

Neumannsche
Randb.

→ siehe

Jackson 1.10

Symmetrie: $G(\vec{r}, \vec{r}') = G(\vec{r}', \vec{r})$ für die Dirichl. Randb.

Green'scher Satz: $\left. \begin{array}{l} \psi(\vec{u}) = G(\vec{u}, \vec{r}) \\ \psi(\vec{u}) = G(\vec{u}, \vec{r}') \end{array} \right\} \begin{array}{l} \psi \Delta_u \psi - \psi \Delta_u \psi = G(\vec{u}, \vec{r}) (-4\pi) \delta(\vec{u} - \vec{r}') - \\ - G(\vec{u}, \vec{r}') (-4\pi) \delta(\vec{u} - \vec{r}) \end{array}$

$$\psi \frac{\partial \psi}{\partial n_u} \Big|_{S_u} = \frac{G(\vec{u}, \vec{r})}{0|_{S_u}} \frac{\partial G(\vec{u}, \vec{r}')}{\partial n_u} \Big|_{S_u} = 0$$

$$\psi \frac{\partial \psi}{\partial n_u} \Big|_{S_u} = \dots = 0$$

$$\begin{aligned} 0 &= -4\pi \int_V d^3u [G(\vec{u}, \vec{r}) \delta(\vec{u} - \vec{r}') - G(\vec{u}, \vec{r}') \delta(\vec{u} - \vec{r})] = \\ &= -4\pi [G(\vec{r}', \vec{r}) - G(\vec{r}, \vec{r}')] \end{aligned}$$

Zum Weiterlesen
Jackson 1.9, 1.10